Mini Project 1: MVPA

Purpose of this assignment:

Related to the second & fourth course objectives (from syllabus):

- Build measured variable path analysis … models with academic rigor,
- Build understanding of SEM research in your field and datasets available to answer questions.

Your goal for this assignment:

- Demonstrate your ability to specify a MVPA model based on theory & RQ in your field.
- Demonstrate your ability to independently run the analysis by adapting code provided in labs
- Demonstrate your ability to curate the relevant outputs to answer your RQ
- Demonstrate your ability to interpret results
- Demonstrate your ability to connect your results to theoretical implications for your field

Out of scope:

- Publication-ready research paper

Submit: Link to a goolge Doc that contains the sections described below. Make sure that you have granted sdevore@hawaii.edu commenter or editor access.

1. Theoretical Background

- Bullet points of the key theoretical background.
- Not a full literature review. Focus on the core theory in your field that drives your RQ.
- 300 word max

2. Research Question

- One (1) question only
- Based on above background
- 100 word max

3. Data

- A brief explanation of the data. Only information relevant to answering your RQ. Not a full description.
- How will you use the data to answer your RQ?
- 300 word max

4. Model Specification

This is a narrowed-down version of the methods section. Note that this should follow from step 1 Theoretical Background.

1. One diagram of the model(s) you will run in this project. If your RQ involves model comparison, then you should submit one diagram per model.
2. The system of equations that the model represents
3. The model-implied covariance matrix representation, developed using the tracing rules

Diagrams & tables only, so no words.

5. Full R code & Outputs

- Generate these in a .qmd or .rmd file, in which you explain what the code is doing. Render as HTML, and submit the output as a separate supplementary file.
- This can contain some EDA and other explorations of the data and doesn't need to be curated.
- No word limit

6. Results

- Only the key outputs necessary to answer the RQ. This will be curated from the the full R code.
- Focus on explaining what the results mean
- Assume no limitations in this section (this section should not include comments like "these resutls are probably due to small sample size, [or any other research design/data/etc problem]"
- Demonstrate that you can identify the key pieces of information needed to answer the RQ and support any claims in the discussion.
- 300 words max, not counting tables/outputs.

7. Discussion

- Answer your RQ based on the outputs.
- Assume there are no limitations when writing this section
- Demonstrate that you can interpret the outputs *as they are*.
- 500 word max

8. Limitations

- This is the only place where you can point out any shortcomings in data/design/etc that make the results & discussion unreliable.
- Bullet points are fine

9. Uncertainties

- As researchers, we often face situations that don't have a difinitive or obvious "correct" approach. What uncertainties did you face? What decision did you make? How did you justify it?
- Bullet points or a table are fine

10. AI use

- Describe if/how you used AI to aid this research. Bullet points are fine.